

Differential Equations Lesson 13

Question 1:

The general solution for the differential equation $y'' + 4y' + 4y = 0$ is $y(x) = C_1 \cdot e^{-x} + C_2 \cdot e^{-3x}$,

where C_1 and C_2 are arbitrary constants.

Taylor series expansion for e^{-x} is $e^{-x} = 1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \frac{(-x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!}$

Taylor series expansion for e^{-3x} is $e^{-3x} = 1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$

The power series solution for the differential equation $y'' + 4y' + 3y = 0$, centered at $x_0 = 0$, is

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 0$ an ordinary point or a singular point?

b) Let $u_n = \frac{(-1)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-1)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

c) Let $u_n = \frac{(-3)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

d) The radius of convergence of the power series solution

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] \text{ is } \underline{\hspace{2cm}} ? \underline{\hspace{2cm}}.$$

e) The interval of convergence of the power series solution

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] \text{ is } \underline{\hspace{2cm}} ? \underline{\hspace{2cm}}.$$

Question 2:

The general solution for the differential equation $y'' + 6y' + 9y = 0$ is $y(x) = C_1 \cdot e^{-3x} + C_2 x \cdot e^{-3x}$,
where C_1 and C_2 are arbitrary constants.

Taylor series expansion for e^{-3x} is $e^{-3x} = 1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$

Taylor series expansion for $x \cdot e^{-3x}$ is $x \cdot e^{-3x} = x \left[1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots \right] = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!}$

The power series solution for the differential equation $y'' + 6y' + 9y = 0$, centered at $x_0 = 0$, is

$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$, where a_0 and a_1 are arbitrary constants.

a) Is the point $x_0 = 0$ an ordinary point or a singular point?

b) Let $u_n = \frac{(-3)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

c) Let $u_n = \frac{(-3)^n \cdot x^{n+1}}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+2}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

d) The radius of convergence of the power series solution $y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$ is $\underline{\hspace{2cm}} ?$

e) The interval of convergence of the power series solution $y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$ is $\underline{\hspace{2cm}} ?$

Question 3:

The power series solution for the differential equation $y'' + 6y' + 5y = 0$, centered at $x_0 = 1$, is

$$y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n, \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 1$ an ordinary point or a singular point?

b) Does the differential equation $y'' + 6y' + 5y = 0$ have any singular points?

c) The radius of convergence of the power series solution $y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n$ is _____?

d) The interval of convergence of the power series solution $y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n$ is _____?

Question 4:

The power series solution for the differential equation $y'' + 16y = 0$, centered at $x_0 = 2$, is

$$y(x) = a_0 \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 16^n}{(2n)!} (x-2)^{2n} \right] + a_1 \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 16^n}{(2n+1)!} (x-2)^{2n+1} \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

- a) Is the point $x_0 = 2$ an ordinary point or a singular point?
- b) Does the differential equation $y'' + 16y = 0$ does not have any singular points?
- c) The radius of convergence of the power series solution $y(x)$ is ____?
- d) The interval of convergence of the power series solution $y(x)$ is ____?

Question 5:

Radius of convergence and interval of convergence of power series solution:

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point.

The power series solution for the differential equation $y'' + 4y' + \frac{1}{x-1} \cdot y = 0$, centered at $x_0 = 3$, is

$$y(x) = a_0 \left[1 - \frac{1}{4}(x-3)^2 + \frac{3}{8}(x-3)^3 - \frac{3}{8}(x-3)^4 + \frac{93}{320}(x-3)^4 \dots \right] + a_1 \left[(x-3) - 2(x-3)^2 + \frac{31}{12}(x-3)^3 - \frac{119}{48}(x-3)^4 \dots \right],$$

where a_0 and a_1 are arbitrary constants.

- a) Is the point $x_0 = 3$ an ordinary point or a singular point?
- b) Is the point $x_0 = 1$ a regular point or a singular point?
- c) The radius of convergence of the power series solution $y(x)$ is ____?
- d) The interval of convergence of the power series solution $y(x)$ is ____?

Question 6:

Radius of convergence and interval of convergence of power series solution:

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point.

The power series solution for the differential equation $y'' + \frac{4}{x^2 - 4} y' + y = 0$, centered at $x_0 = 1$, is

$$y(x) = a_0 \left[1 - \frac{1}{2}(x-1)^2 - \frac{2}{9}(x-1)^3 - \frac{23}{216}(x-1)^4 - \dots \right] + a_1 \left[(x-1) + \frac{2}{3}(x-1)^2 + \frac{5}{18}(x-1)^3 + \frac{2}{9}(x-1)^4 + \dots \right],$$

where a_0 and a_1 are arbitrary constants.

- a) Is the point $x_0 = 1$ an ordinary point or a singular point?
- b) Is the point $x_0 = 2$ an ordinary point or a singular point?
- c) Is the point $x_0 = -2$ an ordinary point or a singular point?
- d) The radius of convergence of the power series solution $y(x)$ is ____?
- e) The interval of convergence of the power series solution $y(x)$ is ____?

Question 7:

The power series solution for the differential equation $y'' + \frac{1}{x^2 + 1}y' + y = 0$, centered at $x_0 = 4$, is

$$y(x) = a_0 \left[1 - \frac{1}{2}(x-4)^2 + \frac{1}{102}(x-4)^3 + \frac{2}{51}(x-4)^4 - \dots \right] \\ + a_1 \left[(x-4) - \frac{1}{34}(x-4)^2 - \frac{140}{867}(x-4)^3 + \frac{9}{2312}(x-4)^4 \dots \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 4$ an ordinary point or a singular point?

b) The singular points for the differential equation $y'' + \frac{1}{x^2 + 1}y' + y = 0$ are ____?

c) The radius of convergence of the power series solution $y(x)$ is ____?

Hint: The distance between two complex numbers, $z_1 = a_1 + b_1 \cdot i$ and $z_2 = a_2 + b_2 \cdot i$, is $d = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2}$.

The distance from $x = 4 = 4 + 0i$ to $x = i = 0 + 1i$ is $\sqrt{(0 - 4)^2 + (1 - 0)^2} = \sqrt{17}$

Question 8:

The power series solution for the differential equation $y'' + 4y' + \frac{1}{x^2 + 9}y = 0$, centered at $x_0 = 1$, is

$$y(x) = a_0 \left[1 - \frac{1}{20}(x-1)^2 + \frac{7}{100}(x-1)^3 - \frac{829}{12000}(x-1)^4 + \dots \right] \\ + a_1 \left[(x-1) - 2(x-1)^2 + \frac{53}{20}(x-1)^3 - \frac{1579}{600}(x-1)^4 + \dots \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 1$ an ordinary point or a singular point?

b) The singular points for the differential equation $y'' + 4y' + \frac{1}{x^2 + 9}y = 0$ are ____?

c) The radius of convergence of the power series solution $y(x)$ is ____?

Hint: The distance from $x = 1 = 1 + 0i$ to $x = 3i = 0 + 3i$ is $\sqrt{(0-1)^2 + (3-0)^2} = \sqrt{10}$

d) The interval of convergence of the power series solution $y(x)$ is ____?

Question 9:

Find the series solution of the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$, centered at the ordinary point $x = 0$.

Substituting $y(x)$, $y'(x)$, and $y''(x)$ into the differential equation $y'' + 4x \cdot y' + y = 0$:

$$\begin{aligned} & \left[2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots \right] + 3x^2 \left(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots \right) + x \left(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \right) = 0 \\ & \left[2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots \right] + \left(3a_1x^2 + 3 \cdot 2a_2x^3 + 3 \cdot 3a_3x^4 + 3 \cdot 4a_4x^5 + 3 \cdot 5a_5x^6 + \dots \right) + \left(a_0x + a_1x^2 + a_2x^3 + a_3x^4 + a_4x^5 + \dots \right) = 0 \end{aligned}$$

Comparing the coefficients of corresponding powers of x on both sides:

$$\text{For } x^0: 2a_2 = 0 \quad \Rightarrow \quad a_2 = 0$$

$$\text{For } x^1: 2 \cdot 3a_3 + a_0 = 0 \quad \Rightarrow \quad a_3 = -\frac{1}{6}a_0$$

$$\text{For } x^2: 3 \cdot 4a_4 + 3 \cdot 1a_1 + a_1 = 0$$

$$\text{For } x^3: 4 \cdot 5a_5 + 3 \cdot 2a_2 + a_2 = 0$$

$$\text{For } x^4: 5 \cdot 6a_6 + 3 \cdot 3a_3 + a_3 = 0$$

$$\text{For } x^5: 6 \cdot 7a_7 + 3 \cdot 4a_4 + a_4 = 0$$

$$\text{For } x^n: (n+1) \cdot (n+2) \cdot a_{n+2} + 3 \cdot (n-1) a_{n-1} + a_{n-1} = (n+1) \cdot (n+2) \cdot a_{n+2} + (3n-2) a_{n-1} = 0 \quad \text{for } n \geq 2$$

a) Recurrence relation is $a_{n+2} = \underline{\hspace{2cm}}?$

b) Express the following terms in terms of a_1 : $a_4 = \underline{\hspace{2cm}}?$, $a_7 = \underline{\hspace{2cm}}?$, $a_{10} = \underline{\hspace{2cm}}?$, $a_{13} = \underline{\hspace{2cm}}?$

c) Express the following terms in terms of a_2 : $a_5 = \underline{\hspace{2cm}}?$, $a_8 = \underline{\hspace{2cm}}?$, $a_{11} = \underline{\hspace{2cm}}?$, $a_{14} = \underline{\hspace{2cm}}?$

d) Express the following terms in terms of a_0 : $a_6 = \underline{\hspace{2cm}}?$, $a_9 = \underline{\hspace{2cm}}?$, $a_{12} = \underline{\hspace{2cm}}?$

e) Power Series Solution for the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$ is

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$y(x) = \underline{\hspace{2cm}}?$$

f) The radius of convergence of the power series solution $y(x)$ is $\underline{\hspace{2cm}}?$.

g) The interval of convergence of the power series solution $y(x)$ is $\underline{\hspace{2cm}}?$.

Question 10:

Find the series solution of the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$, centered at the ordinary point $x = 0$.

Substituting $y(x)$, $y'(x)$, and $y''(x)$ into the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$:

$$(1+x^2)(2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots) + 4x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots) + 4x^2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots) = 0$$

$$\Rightarrow (2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots) + x^2(2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots) + 4x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots) + 4x^2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots) = 0$$

$$\Rightarrow (2a_2 + 2 \cdot 3a_3x + 3 \cdot 4a_4x^2 + 4 \cdot 5a_5x^3 + \dots) + (1 \cdot 2a_2x^2 + 2 \cdot 3a_3x^3 + 3 \cdot 4a_4x^4 + 4 \cdot 5a_5x^5 + \dots) + (4 \cdot 1a_1x + 4 \cdot 2a_2x^2 + 4 \cdot 3a_3x^3 + 4 \cdot 4a_4x^4 + 4 \cdot 5a_5x^5 + \dots) + (4a_0x^2 + 4a_1x^3 + 4a_2x^4 + 4a_3x^5 + 4a_4x^6 + \dots) = 0$$

Comparing the coefficients of corresponding powers of x on both sides:

$$\text{For } x^0: 2a_2 = 0 \quad \Rightarrow \quad a_2 = 0$$

$$\text{For } x^1: 2 \cdot 3a_3 + 4 \cdot 1a_1 = 0 \quad \Rightarrow \quad a_3 = \frac{-4}{6}a_1$$

$$\text{For } x^2: 3 \cdot 4a_4 + 1 \cdot 2a_2 + 4 \cdot 2a_2 + 4 \cdot a_0 = 0$$

$$\text{For } x^3: 4 \cdot 5a_5 + 2 \cdot 3a_3 + 4 \cdot 3a_3 + 4 \cdot a_1 = 0$$

$$\text{For } x^4: 5 \cdot 6a_6 + 3 \cdot 4a_4 + 4 \cdot 4a_4 + 4 \cdot a_2 = 0$$

$$\text{For } x^5: 6 \cdot 7a_7 + 4 \cdot 5a_5 + 4 \cdot 5a_5 + 4 \cdot a_3 = 0$$

$$\text{For } x^n: (n+1) \cdot (n+2) \cdot a_{n+2} + (n-1) \cdot n \cdot a_n + 4 \cdot n \cdot a_n + 4 \cdot a_{n-2} = 0$$

a) Recurring relation is : $a_{n+2} = \underline{\hspace{1cm}}?$

b) Express the following terms in terms of a_0 : $a_4 = \underline{\hspace{1cm}}?$, $a_6 = \underline{\hspace{1cm}}?$, $a_8 = \underline{\hspace{1cm}}?$, $a_{10} = \underline{\hspace{1cm}}?$

c) Express the following terms in terms of a_1 : $a_3 = \underline{\hspace{1cm}}?$, $a_5 = \underline{\hspace{1cm}}?$, $a_7 = \underline{\hspace{1cm}}?$, $a_9 = \underline{\hspace{1cm}}?$

d) Power Series Solution for the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$ is

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$y(x) = \underline{\hspace{1cm}}?$$

e) The radius of convergence of the power series solution $y(x)$ is $\underline{\hspace{1cm}}?$.

f) The interval of convergence of the power series solution $y(x)$ is $\underline{\hspace{1cm}}?$.

